Neural Eavesdropping and the Challenge of Identifying Speakers from sEEG Signals

Ricky Rojas
Department of Data Science
Department of Statistics
Stanford University
rickyro@stanford.edu

Andrew Zhang
Department of Computer Science
Stanford University
andrewxz@stanford.edu

1 Introduction

The advent of deep learning combined with the proliferation of large, high-quality neural datasets has enabled deeper, more complex modeling of the brain. Solving this modeling challenge is key to building brain-computer interfaces (BCIs) which have a multitude of potential uses, from aiding those with paralytic neurological disorders such as ALS (Tang et al. (2023)) to serving as the cornerstone of new technologies in augmented and virtual reality, neuroprosthetics, and cognitive enhancement.

One of the most promising areas of BCI research involves decoding neural activity in naturalistic settings. Unlike controlled experimental paradigms, naturalistic settings present a dynamic and complex array of stimuli that more closely mimic real-world experiences. This approach offers a more holistic understanding of how the brain processes information, contributing to the development of models that generalize better to practical applications.

Our project focuses on this naturalistic decoding challenge using the Brain Treebank repository, which provides an extensive dataset of 43 hours of movie-watching sEEG neural data (Wang et al. (2024)). We give an input of epoched time-series data to an RNN and LSTM and use it to classify the current speaker. This task not only serves as a testbed for decoding speech perception from neural signals but also lays the groundwork for more advanced applications, such as real-time interpretation of neural data in immersive environments or enhancing communication tools for individuals with speech impairments.

Through this project, we aim to bridge the gap between neural signal decoding and practical applications, demonstrating how deep learning techniques can be harnessed to unlock the complexities of the human brain in everyday scenarios.

2 Related Work

Recent intracranial studies have significantly advanced our understanding of how the human brain encodes complex sensory information. For instance, a study in *Nature*, (Leonard et al. (2024)) using Neuropixels probes in the superior temporal gyrus revealed that even within a single cortical column, neurons exhibit highly heterogeneous tuning to diverse speech features such as phonetic cues, prosody, and onsets from silence. Complementarily,Mesgarani et al. (2014) demonstrated that high-density cortical surface recordings in the superior temporal gyrus can capture selective, nonlinear encoding of phonetic features, highlighting the strong auditory connections between speech and neural signals.

In the realm of brain–computer interfaces, Willett et al. (2023) have shown that high-performance speech neuroprostheses are feasible using intracortical microelectrode arrays. Their pipeline—built around a five-layer gated recurrent unit (GRU) network combined with a large-vocabulary language model—achieved impressive decoding performance in real time, even for large vocabularies. Their approach highlights how robust speech decoding can be achieved through spatially intermixed tuning to speech articulators within a small cortical region.

Intracerebral electrophysiological investigations of high-level recognition, such as those reviewed by Rossion et al. (2023),Zhao et al. (2024), and Liu et al. (2021), have provided objective, frequency-domain measures of complex perceptual processes, including face and word recognition. These studies leverage techniques like fast periodic visual stimulation (FPVS) combined with Stereo-

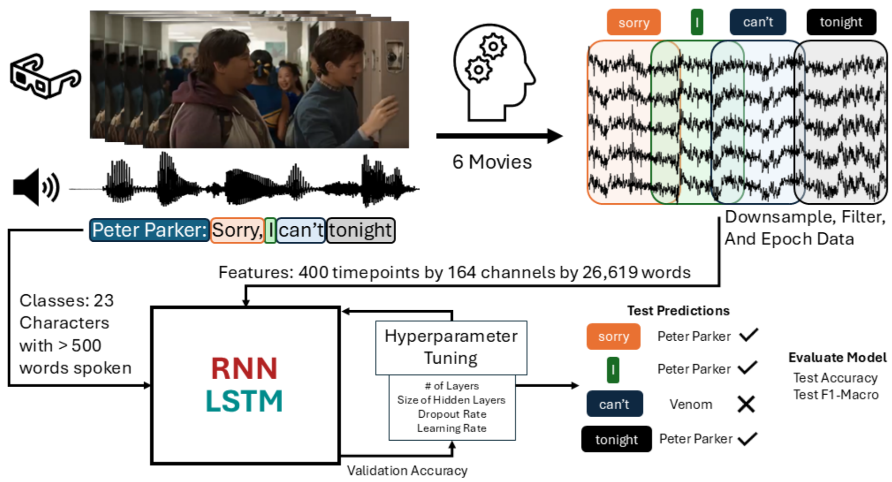

Figure 1: Process Diagram: The audio-visual stimuli is processed by the subject and the resulting neural activity is captured by 164 intracranial electrodes. After processing and restricting the data to characters who spoke over 500 words and split into train/validation/test sets, the train and validation sets are used to tune the hyperparameters for the RNNs and LSTMs. The five models of each model architecture with the highest validation accuracy are then evaluated using the test set.

ElectroEncephaloGraphy (sEEG) to map category-selective responses across the ventral occipitotemporal cortex. Although their primary focus has been on visual recognition, the methodological framework—high temporal and spatial resolution, objective quantification of neural responses, and use of clinical populations—resonates with our work. In our study, we extend these approaches by training recurrent neural network architectures (RNNs and LSTMs) on epoched sEEG data acquired during movie-watching to classify speaker identity. This not only complements prior intracranial findings but also pushes toward more translational BCI applications for decoding naturalistic character processing.

3 Data and Feature Engineering

Our model is trained on data from Subject 2 of the Brain TreeBank as they had the longest watch time of 13.5 hours across 7 movies. The neural data was captured by 164 intracranial electrodes in varying regions of the brain at 2400 Hz. In order to ensure sufficient training data, of the 360 labeled speakers, we reduced the dataset to characters who spoke at least 500 words during the movie. This left us with a total of 23 classes with an average of 1244 words spoken (and a standard deviation of 786 words). Movie 5 (Avenger's Infinity War) had no characters meet our threshold and was excluded. We are unable to conduct a cross-subject analysis because each subject has a different number of electrodes at different locations and watched different movies.

We aligned the onset time of each word with the closest sample from the electrode data and removed words that occurred outside of the sampling period. We then applied notch filters at multiples of 60 Hz to remove line noise and downsampled the data to 200 Hz. Finally, we epoched the data from one second before to one second after word onset to capture the neural response. This resulted in 28,619 word samples of size 164 by 400 (electrodes by timepoints). We generated an 80/10/10 train/validation/test set using stratified shuffle splits so that each set has a similar distribution of classes and used the validation set to tune the hyperparameters.

4 Methods and Model Architecture

4.1 RNN Architecture

The model consists of a multi-layer RNN followed by a fully connected classification layer. The RNN processes time-series data in a batch-first format, meaning input tensors have the shape (batch size, sequence length, feature size). We decided to run batches of 256, so there would be around 16.8 million $(256 \cdot 400 \cdot 164)$ parameters being fed into the neural network. We felt this was a satisfactory amount of parameters to make sure we were within memory limitations. Therefore the input tensors have a shape of (256, 400, 164). The key architectural components include:

Recurrent Layer: A multi-layer RNN. More layers allow for better hierarchical feature extraction. The last hidden state is used as the input to the classifier. The primary hyperparameters we tuned for this layer was the size of the hidden state, and the number of layers. We hypothesized that changing and tuning the hyperparameters in these layers would be the most fruitful in lowering the loss.

Dropout Layer: Dropout is a regularization technique designed to mitigate overfitting by randomly deactivating a subset of neurons during training each epoch. In our model, dropout is applied to the outputs of the recurrent layers before they are fed into the fully connected classifier. This random deactivation forces the network to develop redundant, robust feature representations rather than relying on specific neuron activations, thereby enhancing generalization and stability when processing unseen data (Yadav (2022)).

Activation Function: The recurrent layer uses the tanh activation function, $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, mapping inputs into the range [-1, 1] before sending the output to the next layer. This function helps introduce nonlinearity and regulate the magnitude of our hidden states.

Fully Connected (FC) Layer: A linear transformation that maps the last hidden state to class logits.

Weight Initialization: Our weights and bias initialization were done with strictly zeroes. This is because, in previous trials, random Xavier initialization of the weights did not seem to improve validation or training accuracies compared to zeroes-initialization.

Learning Rate: The learning rate governs the step size during optimization. We used the Adam optimizer, which adapts the learning rate per parameter by computing exponential moving averages of gradients g_t and their squares (Wei (2024)).

Loss Function: The model is trained using categorical cross-entropy loss, which is commonly used for multi-class classification. This function calculates the difference between the predicted class probabilities and the actual labels.

4.2 LSTM Architecture

Long short-term memory (LSTM) architecture is a variation of the RNN architecture, designed to help mitigate the vanishing gradient problem associated with traditional RNNs (Hochreiter and Schmidhuber (1997)). LSTM is also often used for modeling sequential data (Hamad (2023)), which would apply to capturing the time series nature of brain neural activations. All portions of the above RNN architecture description, such as the recurrent layer, dropout layer, activation function, FC layer, and etc. apply to our LSTM architecture as well.

LSTM networks improve on the vanishing gradient problem by introducing a memory cell (C_t) and a set of gating mechanisms that control how information is stored and released. Specifically, an LSTM cell uses three gates: an input gate (i_t) to decide what new information to store, a forget gate (f_t) to determine what information to discard, and an output gate (o_t) to decide what to pass on. The core operations in an LSTM cell, where h_t is the hidden state, x_t is the input, and weight matrixes W, and bias b, are described by:

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f), & C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i), & o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o), \\ \tilde{C}_t &= \tanh(W_C[h_{t-1}, x_t] + b_C), & h_t &= o_t \odot \tanh(C_t). \end{aligned}$$

where σ is the sigmoid function and $\odot$ denotes element-wise multiplication. These gating mechanisms allow the LSTM to selectively remember or forget information over long sequences, effectively mitigating the vanishing gradient issue and enabling the model to capture long-range dependencies in sequential data.

5 Model Results and Analysis

5.1 Hyperparameter Tuning

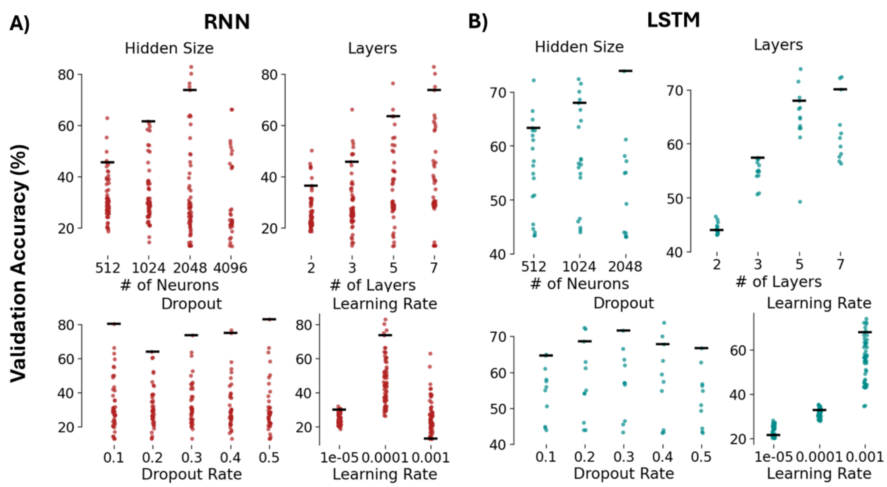

Figure 2: The black lines represent the "best mean model," where the hyperparameters not being plotted are set to the value that yields the best average performance across all models. **(A)** The best mean model is Hidden Size = 2048, Layers = 5, Dropout = 0.4, LR = 0.0001. We ran out of compute before finishing all the models with a hidden size of 4096 (which is why the black bar is missing) but 214 models are plotted. **(B)** The best mean model is Hidden Size = 1024, Layers = 7, Dropout = 0.4, and LR = 0.001. Only the 60 LSTM models corresponding to a learning rate of 0.001 are plotted as the other learning rates led to very poor results.

We hyperparameter-tuned the size of the hidden layers, number of layers, dropout rate, and learning rate using a grid search to 100 epochs, as seen in Figure 2. We tuned the RNNs for 7 days and LSTMs for 11 days on one Nvidia RTX 3090 for each model. Our primary scoring metric was validation accuracy (we also computed the F1, precision, and recall of each model, but in almost all cases, they were comparable to accuracy). It seems that both the RNNs and LSTMs have definitively better learning rates of 0.0001 and 0.001, respectively. Additionally, both models seem to improve as the number of neurons in the hidden layer and number of hidden layers increase. The rate of dropout doesn't seem to make a significant difference to the model, however, from our preliminary results we know that adding any amount of dropout improves our models significantly, with models without dropout performing similarly or worse than baseline.

5.2 Model Results

Our best RNN model has a test accuracy of 80.5% (Table 1) and our best LSTM model has a test accuracy of 72.8% (Table 2). This is significantly above the naive MLE baseline of 9.1% (found by taking the largest class and dividing by the total number of samples). In addition to a higher accuracy, the RNN models trained notably faster.

Table 1: Best RNN Models (LR = 0.0001)

Hidden	Layers	Dropout	Val	Test
2048	7	0.5	**82.8**	**80.5**
2048	7	0.1	80.3	78.7
2048	5	0.4	76.5	74.3
2048	7	0.4	75.1	72.7
2048	7	0.3	73.7	71.8

Table 2: Best LSTM Models (LR = 0.001)

Hidden	Layers	Dropout	Val	Test
2048	5	0.4	**73.9**	**72.8**
1024	7	0.2	72.4	71.3
512	7	0.2	72.2	71.3
1024	5	0.3	71.6	71.2
1024	7	0.1	70.1	67.7

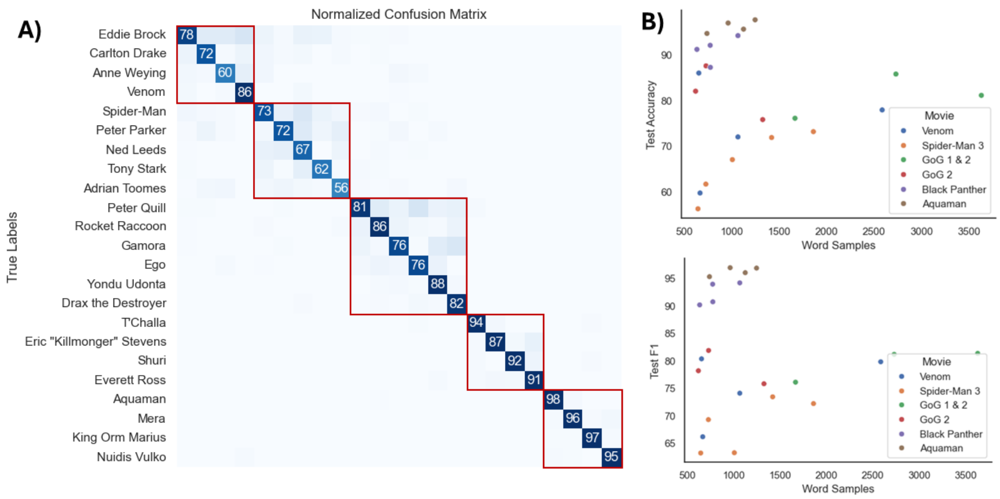

Figure 3: **(A)** A normalized confusion matrix of the predictions from our best RNN model. The diagonal shows the per-class accuracy and red boxes group characters from the same movies together. Guardians of the Galaxy 1 and 2 are grouped together since the notable characters from 1 are also in 2. **(B)** Scatterplots showing test accuracy/F1 vs the number of words spoken by a given character.

5.3 Model Analysis

Looking at the per-class accuracies of our best RNN model we notice two trends (Figure 3A). Firstly, speakers are more likely to be miscategorized as other characters from their movie. There could be a few reasons for this. Our class labels are solely based on the speaker. However, the stimuli are both audio and visual, meaning in scenes with multiple characters on-screen, neural activity correlated with characters other than the speaker may be present. Additionally, some characters, such as Eddie Brock and Venom, are the same person but present differently. This could lead to some confusion in the model, but the effect seems small. Another explanation is that elements unique to each movie (such as loudness or brightness) are present in the neural activity of all the characters from the same movie making it easier to misclassify, however more analysis would be necessary to unravel this effect.

The second trend is that the decoding accuracy/F1 seems to be a function of sample size and class balance (Figure 3B). Movies like Aquaman or Black Panther had a high average accuracy of 96% and 91% respectively with a standard deviation of 200 words spoken between classes. Meanwhile movies like Spider-Man 3 and Venom had lower average accuracies of 66% and 74% with larger standard deviations of 508 and 917 words respectively. Additionally, within these movies, classes with a larger sample size have a significantly higher test accuracy than those with a smaller sample size. We suspect that this is due to using an unweighted cross entropy loss function during backpropagation.

We don't believe our model has overfit to the training data. We used a validation set to perform our hyperparameter tuning and the introduction of dropout is known to combat the issue of overfitting. Additionally, the accuracy of our held-out test set is high.

6 Conclusion

Our analyses have shown that both the RNN and LSTM architectures seem to be well-suited to this task, with the RNN outperforming the LSTM by around 10 percentage points. We believe our models could be further improved by either balancing the classes through random sampling or utilizing a weighted cross-entropy loss function and utilizing more resources to finish hyperparameter tuning, as some hyperparameters seem to be able to be additionally tuned for convergence.

Our results show that it is possible to train a model from very noisy naturalistic neural data to determine the current speaker. Our work expands on previous NLP research to show that RNN architectures can effectively decode signal from multimodal stimuli sources.

7 Contributions

Ricky: Processed and cleaned data. Created the figures and tables. Analyzed model results. Wrote sections 1, 3, 5, and part of 6.

Andrew: Created the model architectures and looked over training and testing. Wrote sections 2, 4, and cowrote part 6.

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